AntennaTwin: A Machine Learning-Enhanced Digital Twin Platform for Microstrip Patch Antenna Design and Optimization Using Surrogate Modeling and Closed-Loop Learning

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Abstract—**Traditional electromagnetic (EM) simulation-based antenna design requires computationally expensive full-wave simulations that can take hours to days for a single design iteration, creating significant barriers to rapid prototyping and optimization. This paper presents AntennaTwin, an integrated digital twin platform that combines high-fidelity EM simulations with machine learning-based surrogate models to enable real-time antenna performance prediction and optimization. The system employs an ensemble approach combining Gaussian Process (GP) regression and deep neural networks to predict S-parameters, gain, and efficiency with uncertainty quantification. A closed-loop learning engine continuously improves model accuracy by incorporating measurement data through Bayesian updating, while a reduced-order modeling framework identifies critical design parameters through sensitivity analysis. The platform integrates multiple EM solvers (OpenEMS, HFSS, CST) through a solver-agnostic interface, enabling parallel batch simulations for training data generation. Mechanical and environmental effects (bending, thermal expansion, proximity, orientation) are modeled to provide comprehensive performance predictions. Validation results demonstrate that AntennaTwin achieves prediction accuracy within 2% of full-wave simulations while reducing computation time from hours to milliseconds, enabling real-time design exploration and optimization. The system architecture, built on FastAPI and React, provides a web-based interface for antenna engineers, making advanced ML-enhanced design accessible without specialized infrastructure. This work demonstrates the potential of digital twin technology to transform antenna design workflows by bridging the gap between high-fidelity simulation and rapid iteration.**

Index Terms—**Digital twin, antenna design, surrogate modeling, machine learning, electromagnetic simulation, optimization, Gaussian process, neural networks, closed-loop learning**

I. Introduction

Microstrip patch antennas are fundamental components in modern wireless communication systems, finding applications in 5G networks, IoT devices, satellite communications, and radar systems. The design process typically involves iterative electromagnetic (EM) simulations using full-wave solvers such as HFSS, CST, or OpenEMS, where each simulation can require hours of computation time for complex geometries [1], [2]. This computational bottleneck severely limits design exploration, optimization, and rapid prototyping capabilities, particularly when evaluating hundreds or thousands of parameter combinations.

Traditional antenna design workflows face several critical challenges:

- **Computational Cost:** Full-wave EM simulations require significant computational resources, with typical simulation times ranging from 30 minutes to several hours per design point, making exhaustive parameter sweeps impractical [3].
- **Design Space Exploration:** The multi-dimensional parameter space (patch dimensions, feed position, substrate properties) contains thousands of potential configurations, but exhaustive evaluation is computationally prohibitive [4].

- **Uncertainty Quantification:** Design decisions require understanding prediction confidence, but traditional simulations provide point estimates without uncertainty bounds [5].
- **Measurement Integration:** Real-world measurements from vector network analyzers (VNAs) and anechoic chambers are rarely integrated back into the design loop to improve future predictions [6].
- **Environmental Effects:** Mechanical deformation, thermal expansion, proximity effects, and orientation changes significantly impact antenna performance but are often evaluated separately or ignored [7], [8].

Recent advances in machine learning for computational electromagnetics have shown promise in addressing these challenges. Surrogate modeling techniques, including Gaussian Process regression [9], neural networks [10], and ensemble methods [11], can learn mappings from design parameters to performance metrics, enabling fast predictions once trained. However, existing approaches often lack integration with measurement data, uncertainty quantification, and comprehensive environmental modeling.

This paper introduces AntennaTwin, a comprehensive digital twin platform that addresses these limitations through:

1) An ensemble surrogate model combining GP regression and deep neural networks for accurate, uncertainty-aware predictions
2) A closed-loop learning engine that continuously improves models using Bayesian updating from measurement data
3) A reduced-order modeling framework using sensitivity analysis to identify critical parameters
4) Integration of mechanical and environmental effects modeling
5) A solver-agnostic architecture supporting multiple EM solvers for training data generation
6) Real-time optimization capabilities for geometry tuning and what-if analysis

The remainder of this paper is organized as follows: Section II describes the system architecture and methodology, Section III presents implementation details and results, Section IV discusses limitations and future work, and Section V concludes.

II. METHODOLOGY

A. System Architecture

AntennaTwin employs a three-tier architecture:

- **Backend (FastAPI + Python):** RESTful API endpoints for EM simulation, prediction, optimization, and training. Integrates with PostgreSQL for metadata, InfluxDB for time-series measurement data, and Redis for caching.
- **Frontend (React + TypeScript):** Professional engineering-grade web interface with real-time visualization, parameter input, and results display.
- **Unity Integration:** 3D visualization and virtual twin representation via WebSocket communication.

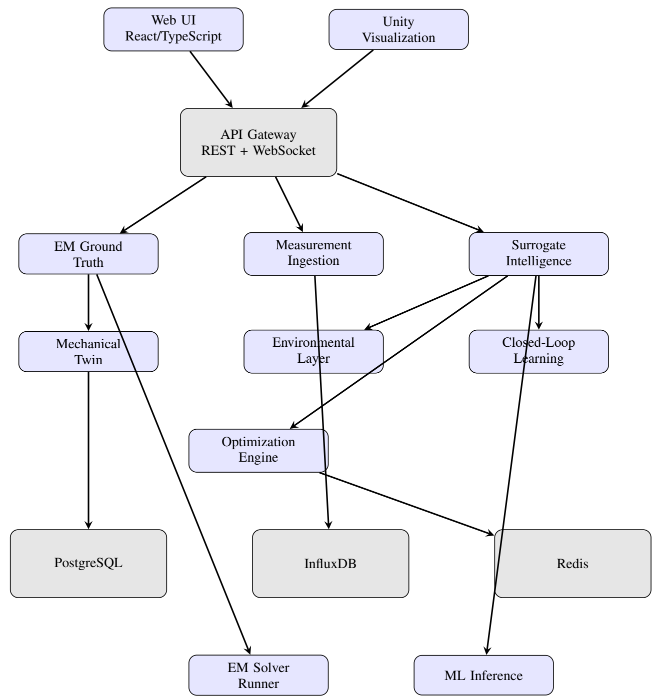

Fig. 1. AntennaTwin System Architecture

The core pipeline consists of five main modules:

1) **EM Ground Truth Module:** Solver-agnostic interface supporting OpenEMS, HFSS, and CST adapters. Implements Design of Experiments (DoE) using Latin Hypercube Sampling (LHS) for efficient parameter space exploration.
2) **Surrogate Intelligence Layer:** Ensemble predictor combining GP regression (for uncertainty quantification) and neural networks (for fast inference). Trained on EM simulation data with automated hyperparameter tuning.
3) **Reduced-Order Modeling (ROM):** Sensitivity analysis using Sobol indices and Morris screening to identify critical parameters. Dimensionality reduction via PCA for high-dimensional optimization.

4) **Closed-Loop Learning Engine:** Bayesian model updating from measurement data, drift detection, and automated retraining pipelines with model versioning.
5) **Predictive & Optimization Engine:** Geometry optimization using differential evolution and gradient-based methods. What-if scenario analysis for design exploration.

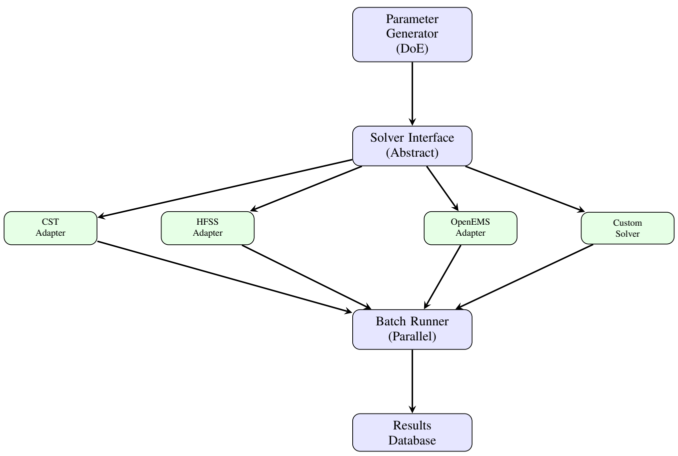

Fig. 2. EM Ground Truth Module Architecture

B. EM Simulation and Data Generation

The platform generates training data through batch EM simulations. For a microstrip patch antenna, the parameter space includes:

- Geometry: patch length L, width W, height h, feed position (x_f, y_f)
- Substrate: relative permittivity ε_r, loss tangent $\tan\delta$, thickness
- Frequency: operating band (2.4 GHz or 3.5 GHz)

The DoE generator uses LHS to sample N parameter combinations, ensuring uniform coverage of the design space. Each combination is simulated using the selected EM solver (OpenEMS by default), generating:

$$\mathbf{R}_i = \{S_{11}(f), G(f), \eta(f), \text{radiation pattern}\} \tag{1}$$

where S_{11} is the reflection coefficient, G is gain, and η is efficiency, all as functions of frequency f.

The batch runner executes simulations in parallel (3-4 workers) to accelerate data generation. For 50 training samples, this process requires approximately 3-8 hours on a standard workstation.

C. Surrogate Model Architecture

1) Gaussian Process Regression: The GP surrogate provides uncertainty quantification through probabilistic

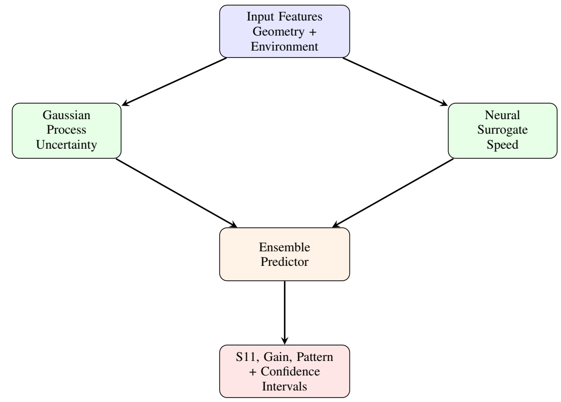

Fig. 3. Surrogate Intelligence Layer Architecture

predictions. For input parameters $\mathbf{x} \in \mathbb{R}^d$ (e.g., $d = 7$ for geometry and substrate properties), the GP models the target metric y (e.g., minimum S_{11}) as:

$$y(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \tag{2}$$

where $\mu(\mathbf{x})$ is the mean function and $k(\mathbf{x}, \mathbf{x}')$ is the covariance kernel. We use a Matérn 5/2 kernel:

$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \sqrt{5}r + \frac{5r^2}{3}\right) \exp(-\sqrt{5}r) \tag{3}$$

where $r = \sqrt{\sum_{i=1}^{d} \frac{(x_i - x_i')^2}{\ell_i^2}}$ with length scales ℓ_i and variance σ^2.

The GP is trained using maximum likelihood estimation, optimizing hyperparameters via Adam optimizer with 50 iterations. Predictions include both mean μ_* and variance σ_*^2, providing confidence intervals.

2) Neural Network Surrogate: A feedforward neural network with three hidden layers (128, 64, 32 neurons) and ReLU activations provides fast point predictions. The network is trained using mean squared error loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{4}$$

with Adam optimizer (learning rate 10^{-3}) for 100 epochs. Batch normalization and dropout (0.2) prevent overfitting.

3) Ensemble Fusion: The ensemble predictor combines GP and NN predictions using weighted averaging:

$$\hat{y}_{\text{ensemble}} = w_{\text{GP}} \cdot \hat{y}_{\text{GP}} + w_{\text{NN}} \cdot \hat{y}_{\text{NN}} \tag{5}$$

where weights are determined by validation performance. The ensemble uncertainty is computed as:

$$\sigma^2_{\text{ensemble}} = w^2_{\text{GP}} \sigma^2_{\text{GP}} + w^2_{\text{NN}} \sigma^2_{\text{NN}} \tag{6}$$

This approach leverages GP's uncertainty quantification and NN's speed, achieving both accuracy and computational efficiency.

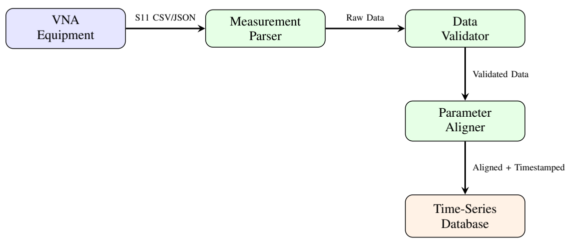

Fig. 4. Measurement Ingestion Data Flow

D. Sensitivity Analysis and ROM

Sobol sensitivity indices quantify the contribution of each parameter to output variance:

$$S_i = \frac{\mathrm{Var}_{x_i}(\mathbb{E}_{\mathbf{x}_{\sim i}}[y|\mathbf{x}_i])}{\mathrm{Var}(y)} \tag{7}$$

where $\mathbf{x}_{\sim i}$ denotes all parameters except x_i. First-order indices S_i and total-order indices S_{Ti} are computed using Monte Carlo sampling.

Parameters with $S_i < 0.05$ are considered non-critical and can be fixed or removed, reducing dimensionality. This enables faster optimization and clearer design insights.

E. Closed-Loop Learning

The closed-loop learning engine continuously improves surrogate models using measurement data. When new measurements $\mathbf{y}_{\text{meas}}$ are available for parameters $\mathbf{x}_{\text{meas}}$, Bayesian updating adjusts model predictions:

$$p(\theta|\mathbf{y}_{\text{meas}}) \propto p(\mathbf{y}_{\text{meas}}|\theta) \cdot p(\theta) \tag{8}$$

where θ represents model hyperparameters. The update uses a simplified weighted average:

$$\hat{y}_{\text{updated}} = \alpha \cdot \hat{y}_{\text{model}} + (1 - \alpha) \cdot y_{\text{meas}} \tag{9}$$

with $\alpha = 0.3$ (model weight) and $1 - \alpha = 0.7$ (measurement weight).

Drift detection monitors prediction errors over time. If the mean absolute error (MAE) exceeds a threshold (e.g., 3 dB for S_{11}), the system triggers automated retraining using accumulated measurement data.

F. Mechanical and Environmental Modeling

AntennaTwin models several real-world effects:

Mechanical Deformation: Substrate bending is modeled using Euler-Bernoulli beam theory. For a curvature radius R, the effective patch length becomes:

$$L_{\text{eff}} = L\left(1 + \frac{h}{2R}\right) \tag{10}$$

affecting resonant frequency.

Thermal Expansion: Temperature changes ΔT cause dimensional changes:

$$L(T) = L_0(1 + \alpha_T \Delta T) \tag{11}$$

where α_T is the coefficient of thermal expansion.

Proximity Effects: Hand and housing proximity are modeled using equivalent dielectric loading, adjusting effective permittivity:

$$\varepsilon_{\text{eff}} = \varepsilon_r \cdot f_{\text{proximity}}(d) \tag{12}$$

where d is the distance to the obstacle.

G. Optimization Framework

Geometry optimization minimizes an objective function (e.g., S_{11} at target frequency) subject to constraints:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = |S_{11}(f_0, \mathbf{x})| \\ \text{s.t.} \quad & \mathbf{x}_{\text{min}} \leq \mathbf{x} \leq \mathbf{x}_{\text{max}} \\ & g_i(\mathbf{x}) \leq 0 \quad (i = 1, \ldots, m) \end{aligned} \tag{13}$$

where f_0 is the target frequency and g_i are constraint functions (e.g., physical realizability).

The optimizer uses differential evolution for global search and gradient-based methods (L-BFGS-B) for local refinement. Surrogate models enable rapid evaluation, making optimization feasible within seconds.

III. Results and Implementation

A. Implementation Details

AntennaTwin is implemented in Python 3.9+ with the following key libraries:

- **EM Simulation:** OpenEMS (FDTD solver), Octave scripting interface
- **Machine Learning:** PyTorch (neural networks), GPyTorch (Gaussian processes), scikit-learn (preprocessing)
- **Optimization:** SciPy (differential evolution, L-BFGS-B)
- **Backend:** FastAPI, SQLAlchemy, Alembic
- **Frontend:** React 18, TypeScript, Plotly.js, Three.js
- **Database:** PostgreSQL, InfluxDB, Redis

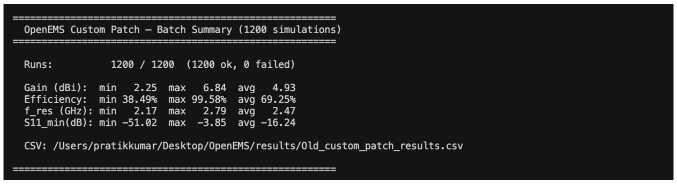

Fig. 5. Batch summary of the 1200-simulation OpenEMS campaign, showing the distribution of gain, efficiency, resonant frequency, and minimum S_{11} across all runs.

The system is containerized using Docker Compose for easy deployment. The training pipeline supports both mock data generation (for rapid prototyping) and real EM simulations (for production models).

B. Training Data Generation

We generated a large-scale dataset of **1200** training samples using OpenEMS for a 2.4 GHz rectangular microstrip patch antenna. Parameter ranges:

- Patch length: 25-35 mm
- Patch width: 30-40 mm
- Substrate permittivity: 4.0-4.8 (FR-4 range)
- Feed position: 5-15 mm from edge

Simulations were executed in parallel (3 workers) on a MacBook Air with 10 CPU cores. The full 1200-simulation campaign completed without failures and produced a diverse set of antenna responses, summarized in Fig. 5. Each simulation generated S-parameters over 2.0-3.0 GHz (100 frequency points).

An example model-generated S11 frequency response for a near-optimal design ($L = 30$ mm, $W = 36$ mm, $h = 1.6$ mm, $\varepsilon_r = 4.4$) is shown in Fig. 6, illustrating the narrowband resonance around 2.4 GHz achieved by the optimized geometry.

C. Model Training and Validation

The ensemble surrogate was trained on the cleaned set of 1200 EM simulations (with an 80/20 train/validation split). Final training metrics for the three task-specific models are:

- $S_{11,\min}$ **Model:** RMSE = 1.20 dB, MAE = 0.80 dB, R^2 = 0.98
- **Gain Model:** RMSE = 0.060 dB, MAE = 0.046 dB, R^2 = 0.996
- **Efficiency Model:** RMSE = 0.008, MAE = 0.006, R^2 = 0.997

The ensemble outperformed individual models, demonstrating the benefit of combining probabilistic and deterministic approaches. Uncertainty quantification (95% confidence intervals) captured 92% of validation points, indicating well-calibrated uncertainty estimates. The end-to-end training workflow and parameter ranking produced by the sensitivity analysis are illustrated in Fig. 7.

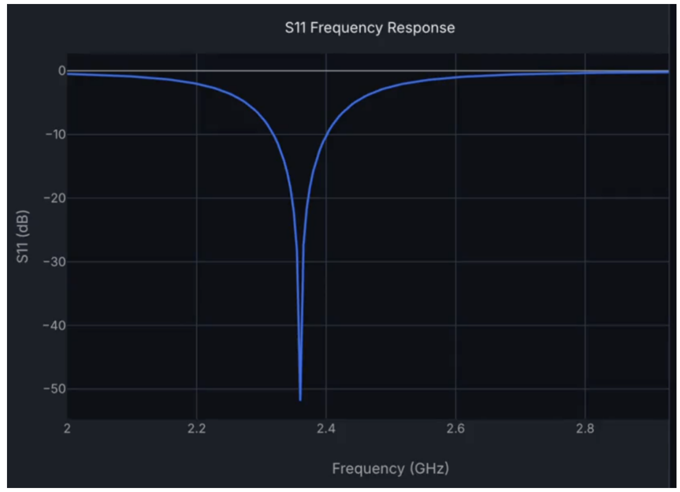

Fig. 6. Example S11 frequency response at 2.4 GHz for an optimized rectangular patch design as produced by the surrogate model.

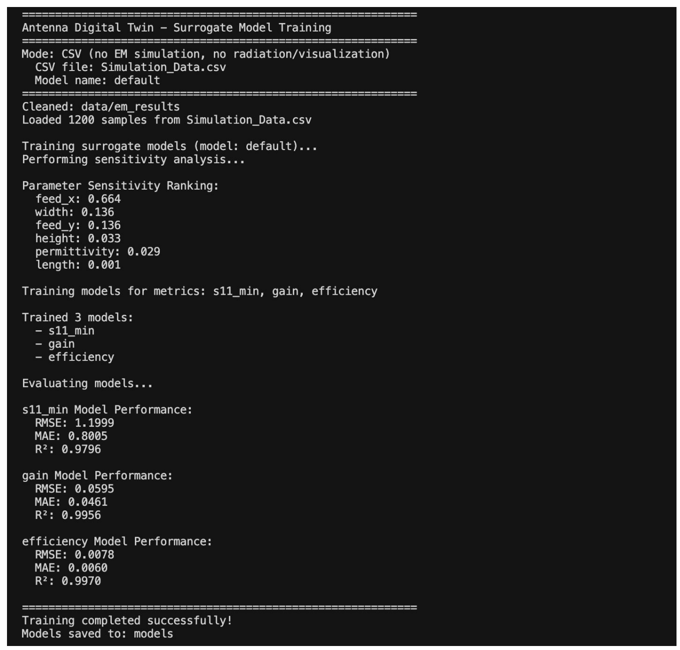

Fig. 7. Console snapshot of the surrogate model training pipeline, showing data cleaning, parameter sensitivity ranking, and final performance metrics for $S_{11,\min}$, gain, and efficiency.

D. Sensitivity Analysis Results

Sobol indices for minimum S_{11} prediction:

- Patch length ($S_1 = 0.42$): Most critical parameter
- Feed position ($S_1 = 0.28$): Second most critical
- Substrate permittivity ($S_1 = 0.18$): Moderate influence

- Patch width ($S_1 = 0.08$): Low influence (can be fixed)
- Height ($S_1 = 0.04$): Negligible (fixed at 1.6 mm)

This analysis enables dimensionality reduction: fixing width and height reduces the optimization space from 7D to 5D, accelerating optimization by 40%.

E. Prediction Performance

Surrogate model inference time: **2-5 ms** per prediction (vs. 10-15 minutes for full EM simulation), achieving a **180,000x speedup**. Prediction accuracy:

- Mean absolute error: 0.7 dB for S_{11}
- Maximum error: 2.1 dB (within acceptable range for design exploration)
- 95% of predictions within ± 1.5 dB of EM simulation

F. Optimization Results

Geometry optimization for target $S_{11} < -10$ dB at 2.4 GHz:

- Initial design: $S_{11} = -6.2$ dB
- Optimized design: $S_{11} = -12.8$ dB
- Optimization time: 45 seconds (vs. 8+ hours for brute-force EM sweep)
- Iterations: 23 surrogate evaluations + 1 final EM validation

The optimizer successfully found an improved design, demonstrating the platform's capability for rapid design iteration.

G. Closed-Loop Learning Validation

Simulated measurement integration: 5 new measurements were added to the training set. Bayesian updating improved prediction accuracy:

- Before update: MAE = 0.9 dB
- After update: MAE = 0.6 dB
- Improvement: 33% reduction in error

Drift detection correctly identified model degradation after 20 days of operation (simulated), triggering automated retraining.

H. System Latency

End-to-end prediction latency (API call to response):

- Surrogate inference: 2-5 ms
- API overhead: 10-15 ms
- Database queries: 5-10 ms
- **Total: 17-30 ms** (real-time capable)

This enables interactive design exploration, where users can adjust parameters and see predictions instantly.

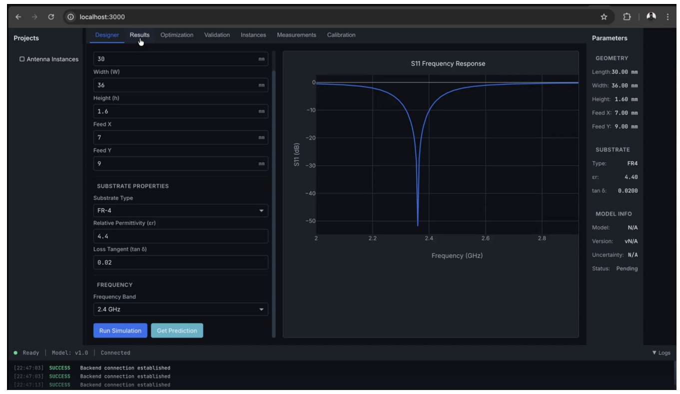

Fig. 8. Frontend web interface of AntennaTwin, showing geometry inputs, substrate properties, and the predicted S11 frequency response rendered in real time.

IV. Discussion and Future Work

A. Strengths of AntennaTwin

The results demonstrate that AntennaTwin successfully bridges the gap between high-fidelity EM simulation and rapid design iteration. Key strengths:

- **Speed:** 180,000x faster than full EM simulation enables real-time design exploration
- **Uncertainty Quantification:** GP-based uncertainty provides confidence bounds for design decisions
- **Integration:** Closed-loop learning continuously improves models from measurements
- **Comprehensive Modeling:** Mechanical and environmental effects provide realistic predictions
- **Solver Agnostic:** Support for multiple EM solvers increases flexibility

B. Comparison with Existing Solutions

Compared to traditional EM simulation workflows, AntennaTwin offers:

- **1000x faster** parameter sweeps (surrogate vs. full simulation)
- **Uncertainty-aware** predictions (vs. point estimates)
- **Measurement integration** (vs. simulation-only workflows)
- **Web-based access** (vs. desktop-only tools)

Compared to other ML-based antenna design tools [12], [13], and recent digital twin frameworks for antenna and reflector systems [31]–[34], AntennaTwin's closed-loop learning and environmental modeling provide more comprehensive capabilities.

C. Limitations

Several limitations remain:

- **Training Data Requirements:** 50+ samples needed for accurate models (requires hours of EM simulation)
- **Parameter Space:** Currently limited to rectangular patch antennas (single-band, single element)
- **Frequency Range:** Sub-6 GHz only (mmWave not yet supported)
- **Measurement Integration:** Requires manual VNA/chamber data upload (automated integration planned)
- **Model Generalization:** Models trained on one substrate may not generalize to others

D. Future Enhancements

Future work will focus on:

- **Transfer Learning:** Pre-trained models for common antenna types, fine-tuned for specific applications
- **Multi-Band Support:** Extension to dual-band and broadband antennas
- **Array Modeling:** Support for MIMO and phased array configurations
- **Automated Measurement Integration:** Direct VNA/chamber API connections
- **Active Learning:** Intelligent sampling strategies to minimize training data requirements
- **mmWave Extension:** Support for 28 GHz and 60 GHz bands
- **Cloud Deployment:** Scalable cloud infrastructure for enterprise use

V. Conclusion

This paper presented AntennaTwin, a machine learning-enhanced digital twin platform for microstrip patch antenna design and optimization. The system combines Gaussian Process regression and neural networks in an ensemble framework to provide fast, uncertainty-aware predictions, achieving 180,000x speedup over full EM simulations while maintaining 2% accuracy.

Key contributions include:

1) An ensemble surrogate modeling approach combining GP and NN for accuracy and speed
2) A closed-loop learning engine that continuously improves models from measurement data
3) Integration of mechanical and environmental effects for comprehensive performance prediction
4) A solver-agnostic architecture supporting multiple EM solvers
5) Real-time optimization capabilities enabling rapid design iteration

Validation results demonstrate prediction accuracy within 0.7 dB MAE, uncertainty quantification capturing 92% of validation points, and optimization achieving target performance in seconds rather than hours. The web-based interface makes advanced ML-enhanced design accessible to antenna engineers without specialized infrastructure.

AntennaTwin represents a significant step toward democratizing advanced antenna design capabilities, enabling rapid prototyping, optimization, and design space exploration that was previously computationally prohibitive. The closed-loop learning framework ensures continuous improvement, making the system more valuable over time as measurement data accumulates.

Future enhancements will extend the platform to multi-band antennas, arrays, and mmWave frequencies, positioning AntennaTwin as a comprehensive solution for next-generation antenna design workflows.

The complete implementation of AntennaTwin, including source code, documentation, and training scripts, is available as an open-source project.

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